MILITARY SPECIFICATION

SEALING COMPOUND, INTEGRAL FUEL TANKS AND FUEL CELL CAVITIES, INTERMITTENT USE TO 360°F (182°C)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for a high temperature resistant fuel tank, filleting and fay surface sealing compound for sealing integral fuel tanks, mold line and fuel tank faying surfaces. The sealant is capable of withstanding long term exposure from -65° to 250°F (-54° to 121°C) and short term exposures up to 360°F (182°C).

1.2 Classification. The sealing compound shall be of the following classes:

Class A - Sealing material, suitable for brush application.

Class B - Sealing material, suitable for application by extrusion gun or spatula.

Class C - Sealing material, suitable for faying surface sealing.

1.2.1 Dash numbers. The following dash numbers shall be used to designate the minimum application time in hours:

Class A dash numbers shall be -1/2, -2, and -4.

Class B dash numbers shall be -1/4, -1/2, -2, -4, and -6.

Class C dash numbers shall be -1/2, -2, -6, and -20.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Air Force Wright Aeronautical Laboratories, MLSA, Wright-Patterson AFB, Ohio 45433 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-250/4	Aluminum Alloy 2024, Plate and Sheet
QQ-A-250/12	Aluminum Alloy 7075, Plate and Sheet
QQ-A-250/13	Aluminum Alloy Alclad 7075, Plate and Sheet
TT-M-261	Methyl Ethyl Ketone, Technical
TT-N-97	Naptha, Aromatic
TT-S-735	Standard Test Fluids: Hydrocarbon
TT-T-548	Toluene, Technical
CCC-C-419	Cloth, Duck, Cotton, Unbleached, Plied Yarns, Army and Numbered
CCC-C-440	Cloth, Cheesecloth, Cotton, Bleached and Unbleached
PPP-B-636	Box, Shipping, Fiberboard
PPP-C-96	Can, Metal, 28 Gage and Lighter
PPP-D-729	Drum, Metal, 55-gallon, For Shipment of Noncorrosive Material
PPP-P-704	Pail, Metal (Shipping, Steel, 1 through 12 Gallon)

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MIL-S-4383	Sealing Compound, Topcoat, Fuel Tank, Buna-N Type
~~MIL-S-5059~~	~~Steel, Corrosion-Resistant (18-8), Plate,~~ Sheet and Strip
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-A-8625	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-T-9046	Titanium and Titanium Alloy, Sheet, Strip and Plate
MIL-C-27725	Coating, Corrosion Preventive, For Aircraft Integral Fuel Tanks
MIL-C-38334	Corrosion Removing Compound, Prepaint, For Aircraft Aluminum Surfaces
MIL-P-38714	Packaging and Packing of Two Components Materials in Semkits

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MIL-C-38736	Compound, Solvent for Use in Integral Fuel Tanks
MIL-C-81706	Chemical Conversion Materials for Coating Aluminum and Aluminum Alloys
MIL-C-87962	Cloth, Cleaning, for Aircraft Fuel Tanks

STANDARDS

FEDERAL

FED-STD-601	Rubber: Sampling and Testing
FED-STD-791	Lubricants, Liquid Fuels, and Related Products; Methods of Testing

MILITARY

MIL-STD-129	Marking and Shipment for Storage
MIL-STD-147	Palletized Unit Loads Validated Oct 1974
MIL-STD-453	Inspection, Radiographic

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1323	Test for Mercaptan Sulfur in Aviation Turbine (Amperometric and Potentiometric Methods)
ASTM D 412	Tension Testing of Vulcanized Rubber

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 Qualifications. The sealing compound furnished under this specification shall be a product which has been tested, and has passed the qualification tests specified herein, and has been listed or approved for listing on the applicable Qualified Products List.

3.2 Materials. The basic ingredient used in the manufacture of the sealing compound shall be synthetic rubber of the polysulfide type. The sealing compound shall cure by the addition of a separate curing compound to the base compound and shall not depend on solvent evaporation for curing. The curing compound shall possess sufficient color contrast to the base compound to permit easy identification of an unmixed or incompletely mixed sealing compound. Neither the base compound nor the cured sealing compound shall be red or pink in color.

3.2.1 Appearance. The base compound and the curing compound (accelerator) shall each be of uniform blend and shall be free of skins, lumps, and gelled or coarse particles. There shall be no separation of ingredients which cannot be readily dispersed.

3.2.2 Mix ratio. The mix ratio of base compound to curing compound shall be 10:1 by weight for all classes and 100:82 by volume for class B.

3.3 Physical properties.

3.3.1 Specific gravity. When tested in accordance with 4.8.1, the specific gravity of the cured sealing compound shall be not greater than 1.65 at standard conditions.

3.3.2 Nonvolatile content. When tested in accordance with 4.8.2, the nonvolatile content of the freshly mixed compound shall be not less than the amount specified below:

Class A - 85 percent by weight.

Class B - 97 percent by weight.

Class C - 92 percent by weight.

3.3.3 Viscosity of base compound. Class A sealing compound shall be suitable for application by brush, and viscosity of the base compound shall be within the range of 100 to 500 poises (10 to 50 Pascal seconds (Pa.s) when tested in accordance with 4.8.3. Class B sealing compound shall be suitable for application by extrusion gun or spatula, and the viscosity of the base compound shall be within the range of 9000 to 16000 poises (900 to 1600 Pa.s) when tested as specified in 4.8.3. Class C sealing compound shall be suitable for application by brush or spatula and viscosity of the base compound shall be within the range of 1000 to 4000 poises (100 to 400 Pa.s) when tested as specified in 4.8.3.

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NOTE: During qualification only, classes B-2, B-4, and B-6 shall be tested for flow, application time, tack-free time, and standard curing rate both immediately after mixing in the Semco Model 1394 mixer, or equal, and after freezing. (See 4.6.4.)

3.3.4 Viscosity of curing compound (catalyst). The viscosity of the curing compound shall be 700 to 1600 poises (70 to 160 Pa.s) when tested in accordance with 4.8.4.

3.3.5 Flow (classes B and C only).

3.3.5.1 Class B. When tested in accordance with 4.8.5.1, freshly mixed class B sealing compound shall exhibit an initial flow within the limits of 0.1 inch (2.5 millimeters (mm)) and 0.75 inch (19.0 mm) when a cylindrical section formed in the flow-test jig is allowed to flow under its own weight on a vertical surface. In addition, the sealing compound shall retain a flow within the limits of 0.1 inch (2.5 mm) and 0.75 inch (19.0 mm) throughout the entire application time.

3.3.5.2 Class C. When tested in accordance with 4.8.5.2, a 0.015 inch (0.40 mm) to 0.020 inch (0.50 mm) layer of fresh sealant shall resist flowing to the extent that at the rated tack-free time the sealant layer shall measure no less than 0.010 inch (0.25 mm).

3.3.6 Application time. When tested in accordance with 4.8.6, the mixed sealing compound shall have an application time of not less than the time in hours specified in Table I after addition of the curing compound, and it shall remain suitable for application during the period. The end of the application time shall be considered as 2500 poises (250 Pa.s) for class A, 15 grams per minute for class B, and 30 grams per minute for class C.

TABLE I. Application time.

Class A	Hours	Class B	Hours	Class C	Hours
		B-1/4	1/4		
A-1/2	1/2	B-1/2	1/2	C-1/2	1/2
A-2	2	B-2	2	C-2	2
A-4	4	B-4	4		
		B-6	6	C-6	6
				C-20	8

3.3.7 Assembly time (Classes C-6 and C-20 only). When tested in accordance with 4.8.7, the mixed sealing compound shall have an assembly time of not less than 6 hours for class C-6 and 20 hours for class C-20. The sealant must squeeze out to a thickness of 0.005 inch (0.13 mm) or less at the bolts.

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3.3.8 Tack-free time. When tested in accordance with 4.8.8, the mixed sealing compound shall cure to a tack-free condition in not more than the time in hours specified in Table II.

3.3.9 Standard curing rate (all classes except C-20). When tested in accordance with 4.8.9, the sealing compound shall have a Type A hardness of not less than 35 after curing under standard conditions (see 4.5.1) for not more than the time specified in Table II.

TABLE II. Tack-free time and curing rate.

Class	Tack-free time (hours)	Curing rate (hours)	Class	Tack-free time (hours)	Curing rate (hours)	Class	Tack-free time (hours)	Curing rate (hours)
			B-1/4	6	16			
A-1/2	10	30	B-1/2	10	30	C-1/2	10	30
			B-2	24	72	C-2	24	*
A-2	24	48	B-4	36	90	C-6	48	*
A-4	*	*	B-6	48	120	C-20	96	NA**

* To be determined.
** NA - not applicable.

3.3.10 Fluid immersed curing rate (classes A-1/2, B-1/4, B-1/2, and C-1/2 only. When tested in accordance with 4.8.10, the sealing compound shall have a hardness of not less than 25 after curing for 48 hours and not less than 35 after curing for 120 hours.

3.3.11 Resistance to thermal rupture. When tested in accordance with 4.8.11, the sealing compound shall retain a pressure of 10 pounds per square inch (psi) with no more than 5/32 inch deformation.

3.3.12 Low-temperature flexibility. When tested in accordance with 4.8.12, the sealing compound shall withstand the low-temperature flexibility test of -67° ± 2°F (-55° ± 1°C) without cracking, checking, or loss of adhesion.

3.3.13 Hydrolytic stability. When tested in accordance with 4.8.13, the sealing compound shall have a Type A hardness of not less than 30.

3.3.14 Shear strength (class C-6 and C-20 only). The shear strength of the cured sealing compound shall be not less than 200 psi (1380 kPa) when tested as specified in 4.8.14. At least 75 percent of the area of separation produced in the shear test specimens shall be within the sealing compound.

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3.3.15 Corrosion. When tested in accordance with 4.8.15, there shall be no corrosion under the sealing compound and the sealing compound shall show no sign of deterioration.

3.3.16. Radiographic density. When tested in accordance with 4.8.16, the H&D density difference between the plate and the plate plus sealant shall not exceed 1.00 as measured by an Ansco-Sweet densitometer, or equal. The density readings through the sealant in the slot shall be approximately 3.00.

3.3.17 Peel strength (all classes except C-20). When tested in accordance with 4.8.17, the sealing compound shall have a minimum peel strength of 20 lbf (89N). All specimens shall exhibit 100 percent cohesive failure, except for bubbles, knife cuts, and other causes that are obviously not the fault of the sealing compound.

3.3.17.1 Peel strength (classes A-1/2, B-1/4, B-1/2, and C-1/2 only). When tested in accordance to 4.8.17.1, the sealing compound shall have a minimum peel strength of 10 lbf (45N) and exhibit 100 percent cohesive failure, except for bubbles, knife cuts, and other causes that are obviously not the fault of the sealing compound.

3.3.18 Resistance to hydrocarbons.

3.3.18.1 Weight loss and flexibility. When tested in accordance with 4.8.18.1, the sealing compound shall lose not more than 8 percent by weight. The specimens shall not crack when bent 180 degrees over an 1/8-inch mandrel.

3.3.18.2 Tensile strength and elongation (class B and C except for class C-20). When class B sealing compound is tested in accordance with 4.8.18.2, the ultimate tensile strength and elongation of the cured compound shall be not less than the values shown in Table III.

TABLE III. Tensile strength and elongation.

Condition	Tensile Strength, psi	Elongation, %
Standard cure	250	250
14-day immersion in jet reference fluid at 140° ± 2°F (60° ± 1°C)	200	200
8 hours at 360° ± 5°F (182° ± 3°C)	200	50

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TABLE III. Tensile strength and elongation (Cont).

Condition	Tensile Strength, psi	Elongation, %
72 hours immersion in jet reference fluid at 140° ± 2°F (60° ± 1°C), followed by 72 hours air-drying at 120° ± 2°F (49° ± 1°C), followed by 7 days at 250° ± 5°F (121° ± 3°C)	250	125
Standard heat cycle in accordance with 3.4.1 followed by 7 days immersion in jet reference fluid at 140° ± 2°F (60° ± 1°C)	100	40

3.3.18.3 Chalking. The sealant shall not chalk when tested in accordance with 4.8.18.3 in jet reference fluid containing no metal ions.

3.3.19 Reparability (all classes except C-20). Sealing compounds shall be suitable for repairing minor breaks in themselves or each other and in all other fuel-aged sealing compound previously qualified under this specification. Specimens of the sealing compound prepared in accordance with 4.8.19 shall have a peel strength of 10 lbf (45N) minimum.

3.3.20 Accelerated storage capability. After storing in accordance with 4.8.20, the base compound shall show no skinning, hardening, separation, nor settling of the material. It shall also conform to the original viscosity specified in 3.3.3. The aged curing compound shall not be adversely affected and shall be capable of being restored by normal agitation to a condition suitable for use. The aged base compound and the aged curing compound, when mixed, shall be in accordance with conditions for flow, application time, assembly time, and tack-free time specified in 3.3.5, 3.3.6, 3.3.7, and 3.3.8, respectively. The peel strength shall be 20 lbf (89N) minimum.

3.3.21 Long term storage. After storing for 9 months in accordance with 4.8.21, the base compound and curing compound shall show no skinning or hardening and shall be capable of being restored by normal agitation to a condition suitable for use. The aged base compound and aged curing compound, when mixed, shall meet the minimum application time, maximum tack-free time, and maximum cure time requirements of Tables I and IV.

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TABLE IV. Tack-free time and curing rate for materials stored for 9 months.

Class	Tack-free Time (hours)	Curing Rate (hours)	Class	Tack-free Time (hours)	Curing Rate (hours)	Class	Tack-free Time (hours)	Curing Rate (hours)
			B-1/4	16	24			
A-1/2	20	40	B-1/2	20	40	C-1/2	20	*
			B-2	48	72	C-2	48	*
A-2	48	72	B-4	72	114	C-6	96	*
A-4	*	*	B-6	96	144	C-20	*	NA

* To be determined.
NA - not applicable.

3.3.22 Hardness. When tested in accordance with 4.8.22, the sealing compound shall have a Type A hardness of not less than 40.

3.3.23 Source inspection. All materials procured by the Government under this specification must be source inspected. (See 4.1.1).

3.4 Workmanship. The workmanship shall be in accordance with high-grade manufacturing practice covering this type of material. The material shall be suitable for its intended use and free of defects which may affect its functionability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Source inspection. Materials procured by the Government under this specification must be source inspected so that there is assurance that the material meets the quality conformance tests (4.4.3) when it leaves the manufacturer's plant. Note that 4.4.2 requires that the samples for quality conformance tests be packaged and mixed as much as practical in the same containers that are being procured.

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4.1.2 Date of manufacture. The date of manufacture is defined as the date the quality conformance tests are completed by the sealant manufacturer.

4.1.3 Shelf-life surveillance and updating.

4.1.3.1 Sampling. The minimum number of samples to be tested during shelf-life surveillance and updating are as follows:

Items in stock	Samples to be tested
up to 100	3
100 to 500	5
more than 500	7

4.1.3.2 Shelf-life surveillance and updating tests. The following inspections are to be conducted for shelf-life surveillance and updating:

a. Condition of container
b. Application time
c. Tack-free time
d. Standard curing rate
e. Viscosity of base compound (not possible with Semkits)
f. Viscosity of curing compound (not possible with Semkits)
g. Peel strength 2 aluminum panels coated with material conforming to MIL-C-27725, age in jet reference fluid 7 days at 140°F (60°C)

Tests are to be conducted in accordance with test methods outlined in this specification for quality conformance tests (see 4.4.3). If the tests are being performed at the end of the stated shelf-life to update the shelf-life of the sealing material, and all tests are passed, the shelf-life will be extended an additional three months. Up to three updatings shall be allowed.

4.1.4 Curing compound (catalyst) replacement. There are instances where sealant will become overaged and require excessive periods of time to cure. Usually, the curing compound will be the component that has deteriorated. It is possible to replace the curing compound so that the sealant can be used. Anytime the curing compound is replaced, all the quality conformance tests (see paragraph 4.4.3) must be met by the final curing compound/base combination.

4.2 Classification of tests. The inspection and testing of the sealing compound shall be classified as follows:

a. Qualification tests (4.3).
b. Quality conformance inspection (4.4).

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4.3 Qualification tests. The first material that a manufacturer must qualify is class B-2. That material must meet all the requirements of this specification with the exception of those requirements that are for other classes. Once qualification for class B-2 is obtained, other classes may be qualified. The formulation for class A and C materials and other class B materials shall be the same as that for the class B-2, except for minor variations necessary for viscosity and application life. It will not always be necessary for the qualifying agency to conduct all the tests on the other classes. In general, the quality conformance tests will be sufficient, although additional tests can be required. The manufacturer will have to show proof that all requirements are met prior to requesting qualification approval for any class. This includes assurance that the material will cure at standard conditions. (Quality conformance testing is conducted on material which has been cured at 140°F (60°C). After the compound has been accepted for qualification, approval will be granted and the material will be identified by reference to the manufacturer's code or formula number.

4.3.1 Qualification test samples. The qualification test samples shall consist of 3 gallons and 2 quarts of the compound upon which qualification is desired. The qualification samples shall be identified as specified herein and forwarded to the activity responsible for qualification as designated in the letter of authorization from that activity (see 6.3).

Samples for qualifications tests:

SEALING COMPOUND, INTEGRAL FUEL TANKS AND FUEL CELL CAVITIES, INTERMITTENT USE TO 360°F (182°C)
Class and dash No.
Specification MIL-S-83430A
Manufacturer's code No.
Name of manufacturer
Submitted by (name) (date) for qualification tests in accordance with MIL-S-83430A under authorization (reference authorizing letter)
Mixing instruction

4.4 Quality conformance inspection.

4.4.1 Batch. A batch shall be the quantity of material run through a mill or mixer at one time.

4.4.2 Sampling for quality conformance tests. An approximate 1-quart sample shall be selected from each batch. The material shall be packaged and mixed as much as practical in the same containers that are being procured; that is, if the procurement is for packaging in accordance with MIL-P-38714 the conformance tests shall be conducted with material mixed in these packages.

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4.4.3 Quality conformance tests. Quality conformance tests for acceptance of individual batches shall consist of the following tests:

a.	Nonvolatile content	4.8.2
b.	Viscosity of base compound	4.8.3
c.	Viscosity of curing compound	4.8.4
d.	Flow (classes B and C only)	4.8.5
e.	Application time	4.8.6
f.	Assembly time (classes C-6 and C-20 only)	4.8.7
g.	Tack-free time	4.8.8
h.	Standard curing rate (all classes except C-20)	4.8.9
i.	Fluid immersed curing rate (class A-1/2, B-1/2, and C-1/2 only)	4.8.10
j.	Resistance to thermal rupture 1/	4.8.11
k.	Shear strength (class C-6 and C-20 only) 1/	4.8.14
l.	Peel strength (all classes except C-20) (4 aluminum panels coated with material conforming to MIL-C-27725, see Table IV. 2 panels in JRF and 2 panels in JRF/Salt Water for 7 days at 140°F (60°C). 1/ Do not use PR-148 Adhesion Promoter.	4.8.17
m.	Peel strength (classes A-1/2, B-1/2, and C-1/2 only). Do not use PR-148 Adhesion Promoter.	4.8.17.1
n.	Weight loss and flexibility 1/	4.8.18.1
o.	Chalking 1/	4.8.18.3
p.	Hardness 1/	4.8.22

1/ In lieu of the 14-day cure specified, specimens shall be subjected to an accelerated cure of 48 hours at 77°F (25°C) relative humidity of 50 percent followed by 24 hours at 140°F (60°C).

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4.5 Standard conditions.

4.5.1 Test conditions. Standard laboratory testing conditions shall be 77° ± 2°F (25° ± 1°C) and 50 ± 5 percent relative humidity. Except as otherwise specified hereinafter, all test specimens shall be prepared, cured, and tested under these conditions.

4.5.2 Standard heat cycle. Unless otherwise specified herein, the standard heat cycle shall consist of 24 ± 1 hour at 260° ± 2°F (127° ± 1°C), 4 ± 1/2 hour at 320° ± 2°F (160° ± 1°C), and 6 ± 1 hour at 358° ± 2°F (181° ± 1°C) divided evenly into 6 portions, each consisting of 4 ± 1/2 hour at 260° ± 2°F (127° ± 1°C), followed by 40 ± 5 minutes at 320° ± 2°F (160° ± 1°C) and 1 hour ± 5 minutes at 358° ± 2°F (181° ± 1°C). At the completion of each 360°F (182°C) exposure, the temperatures shall be reduced to less than 100°F (38°C) before again exposing to the 260°F (127°C) cycle.

4.6 Preparation of test specimens.

4.6.1 Cleaning of test panels. All test panels shall be cleaned by scrubbing and rinsing using MIL-C-38736 solvent and clean MIL-C-87962 Type II gauze which is free of sizing or any other contaminant. The panels will then be wiped dry immediately with gauze. NOTE: All coatings are to be cleaned as described above before the sealant is applied, with one exception. Do not clean after the PR-148 adhesion promoter is applied. Doing so will remove the adhesion promoter.

4.6.2 PR-148 adhesion promoter. When specified, the panel surface shall be treated with PR-148 adhesion promoter. This shall be done immediately after the panel has been cleaned (4.6.1). The PR-148 shall be applied by wetting a clean gauze with PR-148 and wiping the surface. Allow the PR-148 to air dry at least 15 minutes but not more than 2 hours before applying the sealant. If more than 2 hours have elapsed, re-clean and re-apply the PR-148 before applying sealant.

4.6.3 Preparation of sealing compound

4.6.3.1 Qualification tests. The quantity of sealing compound required for the test shall be mixed as thoroughly as practicable. A Semco Model 1394 mixer, or equal, shall be used for class B-2, B-4, and B-6. Class B-1/4, B-1/2, and all classes A and C shall be hand mixed. The mixed sealing compound shall have a minimum inclusion of air. Where applicable the sealing compound, immediately after mixing, shall be placed in cartridges for extrusion from the Semco No. 250 gun, or equal. Sealing compound in sectional-type containers (see 5.1.1.2) are to be machine mixed using a Semco Model 285 mixer, or equal.

4.6.3.2 Quality conformance tests. The sealing compound shall be hand mixed for all classes. MIL-P-38714 containers shall be used when applicable. (See 4.4.2).

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4.6.4 Quick freezing, class B-2, B-4, and B-6, Material Qualifications Tests. After mixing on the Semco Model 1394 mixer, or equal, two cartridges shall be held at room temperature. One of these shall be used for testing application time and the other for tack-free time, curing rate, and flow. The remainder of the cartridges shall be quick frozen. After the materials are loaded into the cartridges, both ends of the cartridge shall be closed off immediately after filling. The installed plunger constitutes a satisfactory plug at one end. The sealant shall be quick frozen immediately in a TT-N-97, Type I, Grade B aromatic naptha and dry ice bath at -80°F (-52°C) or lower for 30 minutes. The cartridge shall be placed in a plastic bag and immersed with its plugged nozzle end down and the upper end about one inch above the liquid level. For tests on tack-free, curing rate, application time, and flow, the class B-2 and B-4 material shall be stored at or below -40°F (-40°C) for at least 16 hours but not more than 48 hours. When frozen sealant stored at -40°F (-40°C) is used for tests, it shall be further conditioned for 2 hours at -67° ± 2°F (-55° ± 1°C) prior to the start of thawout. Thawout shall be accomplished by immersion of the frozen cartridge in a 120° ± 2°F (49° ± 1°C) water bath for 18 minutes with the plugs installed and the upper end of the cartridge one inch above the liquid level. Time zero shall be considered as occurring at the end of the 18 minute period and the timed tests begun. For all other tests, the storage time of the frozen material shall not exceed 10 days at -67° ± 2°F (-55° ± 1°C). Thawout shall be accomplished as above except sealant may be thawed to room temperature if desired.

4.6.5 Application of sealing compound. Unless otherwise specified herein, test panels shall be given an application of sealing compound to produce a coating having a thickness of 1/8 ± 1/64 inch (3.2 ± 0.4mm) when cured. Class A materials shall be built up by 3 applications, allowing a period of time equal to the application life between applications to permit release of solvents. Unless otherwise specified, test specimens shall receive a 14-day cure at 77° ± 2°F (25° ± 1°C) and 50 ± 5 percent relative humidity. Tests on the cured sealing compound shall commence not later than 2 days after completion of the specified cure.

4.6.6 Chemical conversion coating application.

4.6.6.1 Coating preparation. A chemical conversion coating conforming to MIL-C-81706, Class IA, Form II, Method C shall be used. It shall be prepared according to the manufacturer's instructions. The PH of the resulting solution is adjusted to 1.5 using nitric acid.

4.6.6.2 Panel preparation.

a. Vapor degrease using trichloroethane or solvent degrease using trichloroethane or ketone.

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b. Alkaline detergent clean using MIL-C-25769 material or an equivalent commercially available alkaline cleaner. The cleaning may be accomplished by brushing, swabbing or soaking the panels in the detergent solution or by a combination of the above techniques. Rinse the cleaned panels in warm flowing tap water 60°F to 100°F (16°C to 38°C), and check for cleanliness by observing for waterbreak free surface. If a waterbreak occurs on the panel surfaces, return them to the detergent solution and repeat cleaning procedure until a waterbreak free surface is obtained.

c. Immediately transfer the cleaned panels to a deoxidizing solution consisting of the following:

Butyl alcohol - 35 percent by weight
Distilled or deionized water - 22 percent by weight
Isopropyl alcohol - 25 percent by weight
H_3PO_4 (85 percent by weight) - 18 percent by weight

Acid deoxidizer conforming to MIL-C-38334 may also be used. Allow the panels to remain in the above solution for 3 to 5 minutes. Rinse the panels thoroughly under flowing tap water.

4.6.6.3 Coating application (immersion). Transfer the deoxidized panels immediately to the chemical conversion coating solution conforming to MIL-C-81706. Immerse the panels in the solution at standard temperature for 3 to 5 minutes or until a light straw color develops. (Color development time will vary with aluminum alloy being conversion coated). After removal from the conversion coating solution, immediately rinse thoroughly in flowing distilled or deionized water. Arrange the panels in an upright position to permit them to drain dry. Apply test materials to the conversion coated surfaces within 48 hours.

NOTE: Mix the conversion coating solution in either 18-8 stainless steel, polyethylene or other compatible plastic containers. DO NOT MIX IN GLASS CONTAINERS.

4.7 Formulation of jet reference test fluid. The jet reference fluid required for conducting fluid immersion tests shall be formulated as follows:

Toluene (TT-T-548)	30 volumes
Cyclohexane (technical grade)	60 volumes
Iso-octane (TT-S-735, Type 1)	10 volumes
1/Tertiary dibutyl disulfide	1 volume
Tertiary butyl mercaptan	0.015 + 0.0015 weight percent of other four components

1/ The tertiary butyl disulfide shall be doctor sweet.

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When tested for mercaptan and total sulfur in accordance with ASTM D1323 and Method 5201 of FED-STD-791, respectively, the reference fluid shall have the following properties:

a. Total sulfur content - 0.400 ± 0.005 weight percent

b. Mercaptan sulfur content - 0.0050 ± 0.0005 weight percent.

The fluid shall be stored out of contact with light in containers which are inert to the fluid ingredients. (Welded aluminum, non-galvanized welded steel, or glass containers are suitable.) If not used within 90 days after blending, the fluid should be retested for mercaptan and total sulfur content. There shall be no metal ions present in the fluid.

4.8 Test methods.

4.8.1 Specific gravity. Three specimens approximately 1/8 by 1 by 1 inch (3.2 by 25.4 by 25.4 mm) in size shall be cut with a sharp razor blade from a sheet of the sealing compound that has been cured for 14 days. The specimens shall be weighed accurately to 3 decimal places in grams after which they shall be dipped in methyl alcohol and then, while wet, immediately suspended in distilled water at standard conditions and reweighed. The specific gravity shall be computed from the following formula:

$$\frac{\text{Weight in air}}{\text{Weight in air minus weight in water}} = \text{Specific gravity} \qquad (1)$$

4.8.2 Nonvolatile content. Five to 10 grams of freshly mixed sealing compound shall be transferred as rapidly as possible to a previously dried (158° ± 2°F (70° ± 1°C)) and desiccator cooled cup approximately 3 inches (76 mm) in diameter and 3/4 inch (19 mm) in depth. A fitted cover shall immediately be placed over the cup and the weight accurately determined, using an enclosed beam balance that will weigh accurately to the nearest milligram. The weight of the cup and cover shall be determined accurately prior to using, and subtracted from the initial and final weights in order to calculate the new sample weights. The cover shall then be removed, and the sealing compound heated in an oven for 7 days at 158° ± 2°F (70° ± 1°C). It shall then be cooled in a desiccator, the cover replaced, and the weight determined accurately. The percentage of total nonvolatile content shall be calculated as follows:

$$\text{Percent of nonvolatile content} = \frac{\text{Final weight}}{\text{Initial weight}} \times 100 \qquad (2)$$

Three samples shall be used in this determination.

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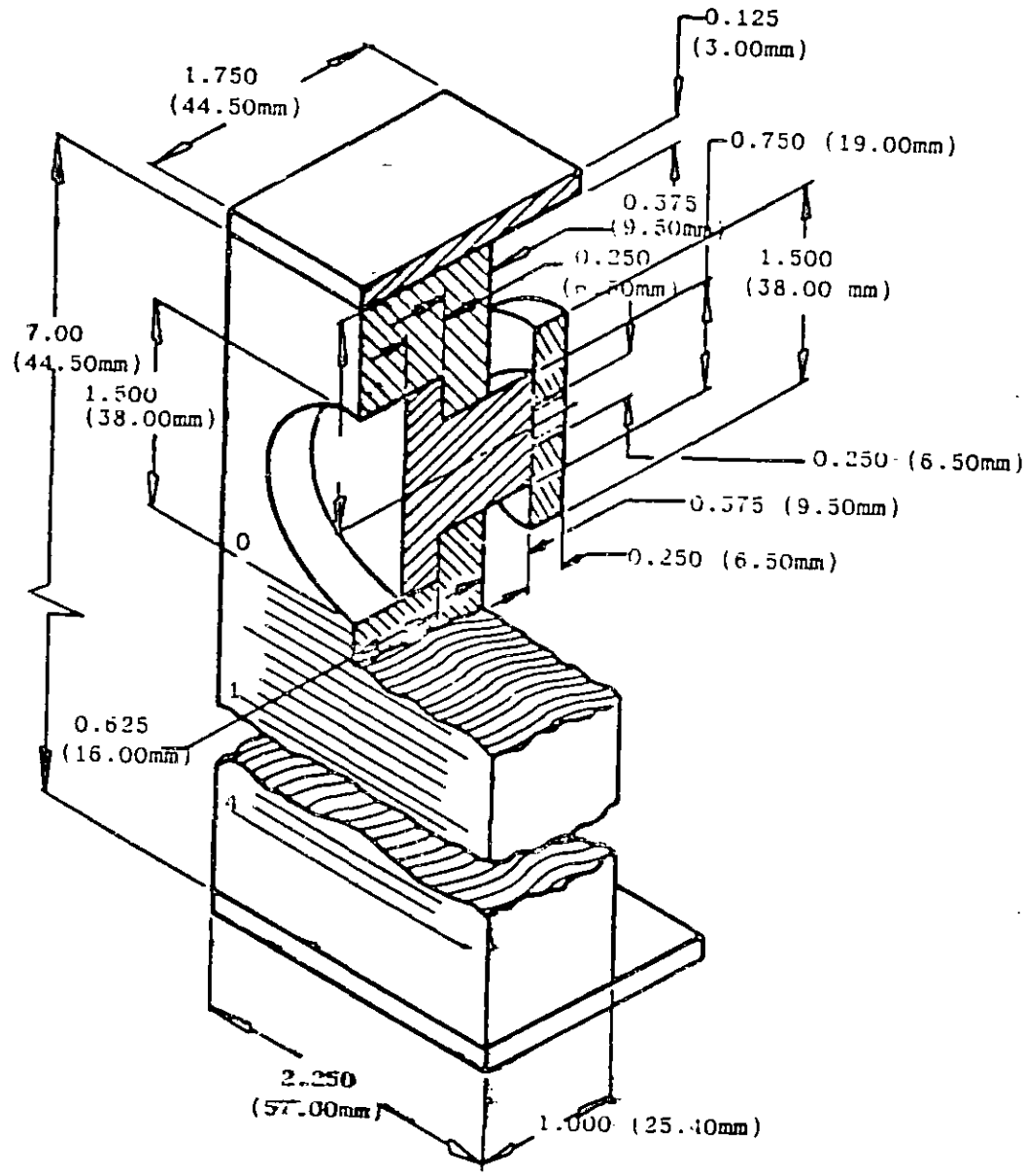
4.8.3 Viscosity of base compound. The viscosity shall be determined with base compound (see 3.3.3) placed in a 1-quart (1-liter) can. The can shall be filled with base compound to within 1/2 inch (10 mm) of the top; covered and stored at room temperature for at least 8 hours. The base compound shall then be thoroughly mixed by stirring slowly for 3 minutes after which the can shall be closed and the material allowed to stand for 1 hour. The Brookfield Model RVF viscosimeter, or equal, shall be used and the reading obtained converted to poises. For class A base materials, the No. 6 spindle at 10 revolutions per minute (rpm) shall be used; for class B material, the No. 7 spindle at 2 rpm shall be used; for class C material, the No. 6 spindle at 2 rpm shall be used. The highest reading shall be taken after the instrument has run in the material for 1 minute.

4.8.4 Viscosity of curing compound (catalyst). The viscosity of the curing compound shall be determined in accordance with 4.8.3 except a No. 7 spindle at 10 revolutions per minute shall be used.

4.8.5 Flow (class B and C only).

4.8.5.1 Class B. A standard Semco sealant gun cartridge, or equal, fitted with a nozzle, shall be filled with freshly mixed class B sealing compound. The gun and sealing compound shall be maintained at standard conditions throughout the test. The test shall be conducted with a flow-test jig as shown in Figure 1 and under flow conditions specified in 3.3.5.1. Depth of plunger tolerance is critical and shall be controlled within the tolerance during all tests. The flow-test jig shall be placed on a table with the front face upward and the plunger depressed to the limit of its travel. Within 15 minutes after the beginning of mixing, enough of the mixed sealing compound shall be extruded from the application gun to fill the recessed cavity of the jig and leveled off even with the block. The test at this interval shall be considered the initial flow of the sealing compound. Within 10 seconds after the leveling operation, the jig shall be placed on its end and the plunger immediately advanced to the limit of its forward travel. The flow measurement shall be taken exactly 30 minutes after the sealing compound has been applied to the test jig. The flow shall be measured from tangent to the lower edge of the plunger to the farthest point to which flow has advanced. As the sealing compound progresses in its application time, the flow-test shall be repeated at the time intervals specified in Table V. All time intervals, other than for the initial test, shall be measured from the end of the mixing period.

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MATERIAL: ALUMINUM ALLOY

DIMENSIONS IN INCHES (MILLIMETERS). UNLESS OTHERWISE SPECIFIED, TOLERANCES:
DECIMALS ± 0.016 INCH (± 0.40 MM)

FIGURE 1. Flow-test jig.

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TABLE V. Flow-test intervals.

Minimum specified application time	Intervals at which flow tests shall be conducted
1/4 hour----------	Initial reading only.
1/2 hour----------	Initial reading only.
2 hours----------	Initial. 50 minutes. 90 minutes.
4 hours----------	Initial. 2 hours. 3-1/2 hours.
6 hours----------	Initial. 3 hours. 5-1/2 hours.

4.8.5.2 Class C. A 0.015 to 0.020 inch (0.40 to 0.50 mm) layer of freshly mixed sealant shall be applied to a QQ-A-250/13 T6 aluminum panel measuring 0.040 by 2-3/4 by 6 inches (1.00 by 70 by 150 mm). Immediately place the panel in a vertical position and allow to stand for a period equivalent to the rated tack-free time (see 3.3.8). Measure the sealant for compliance with 3.3.5.2.

4.8.6 Application time.

4.8.6.1 Class A sealing compound. The base compound and accelerator shall be stabilized at standard conditions for at least 8 hours before a sample of base compound is mixed with the proper amount of accelerator sufficient to fill a standard 1/2-pint (1/4 liter) can (2-7/8 inches) (73 mm) in diameter by 2-7/8 inch (73 mm) high to within 1/2 inch (10 mm) of the top. This can shall be tightly covered except when testing the viscosity. At the end of the rated application time, measured from the beginning of the mixing period, the sealing compound shall be tested for biscosity using a Brookfield Model RVF viscosimeter, or equal, with a No. 7 spindle at 10 rpm. One reading shall be taken after the instrument has run in the material for 1 minute.

4.8.6.2 Class B and class C sealing compound. The base compound, accelerator, and application gun shall be stabilized at standard conditions for at least 8 hours before a minimum of 250 grams of base compound are

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mixed with the proper amount of accelerator. The mixed sealing compound shall be promptly used to fill a standard Semco sealing compound gun cartridge, or equal, having a Semco 440 nozzle with an orifice diameter of 0.125 inch ± 0.005 inch (3.2 ± 0.13 mm). The gun and sealing compound shall be maintained at standard conditions throughout the test. The gun shall be attached to a constant air supply of 90 ± 5 psig (4.3 ± 0.2 kPa). From 2 to 3 inches (5 to 7-1/2 cm) of sealing compound shall be extruded initially to clear trapped air. At the end of the rated application time, measured from the beginning of the mixing period, the sealing compound shall be extruded onto a suitable receptacle for one minute and the amount of extruded sealing compound determined.

4.8.7 Assembly time (class C-6 and C-20 only). Six test panels 0.40 by 1-1/2 by 4 inches (1.00 by 38 by 100 mm) in size shall be prepared from aluminum alloy conforming to temper T6 of QQ-A-250/13. Drill two holes with a number 11 drill, 1.2 inches (30 mm) from one end with centers 3/4 inch (20 mm) apart and 3/8 inch (10 mm) from each side. Deburr and clean with MIL-C-38736 cleaner. Accurately determine the thickness of the panels around the holes. Apply approximately 0.015 inches (0.40 mm) of freshly mixed sealant to the drilled end of three specimens and allow to cure for 1/2 hour. Place the other cleaned panels on those with sealant so that the holes line up and results in a one-inch (25 mm) overlap. Sealant shall cover the entire one-inch (25 mm) faying surface overlap area. Insert two steel bolts (10-32) that have been heat treated to at least 160,000 psi (1.10 Megapascals (MPa)) into the holes and tighten nuts (NAS 679-A3) only until sealant starts to squeeze out. the thickness of the assembly shall be measured at this time and the thickness of the sealant shall be 0.010 to 0.015 inch (0.25 to 0.40 mm). Allow the specimens to be exposed to standard conditions for 6 hours for the class C-6 material and 20 hours for the class C-20 material. Tighten nuts to a torque value of 40 inch lbs (4.5 Newton meter (N.m). Measure the thickness of the assembly at the bolts with a micrometer and from this thickness subtract the thickness of the panels. Evaluate in accordance with 3.3.7.

4.8.8 Tack-free time. An aluminum test panel conforming to temper T6 of QQ-A-250/13 and measuring 0.040 by 2-3/4 by 6 inches (1.00 by 70 by 150 mm) in size shall be cleaned in accordance with 4.6.1 and covered with freshly mixed sealing compound to a depth of 0.040 inch ± 0.010 inch for class A and 0.125 ± 0.015 inch for class B and class C. The sealing compound shall be allowed to cure at standard conditions. At the end of the specified tack-free time, two 1 inch by 6 inch pieces of polyethylene film 0.004 ± 0.002 inch thick shall be applied to the sealing compound using light finger pressure. Two minutes after applying the polyethylene film, the strips shall be slowly and evenly withdrawn at right angles to the sealing compound surface. The polyethylene shall come away clean and free of sealing compound.

4.8.9 Standard curing rate (all classes except C-20). The instantaneous hardness shall be determined in accordance with method 3021 of

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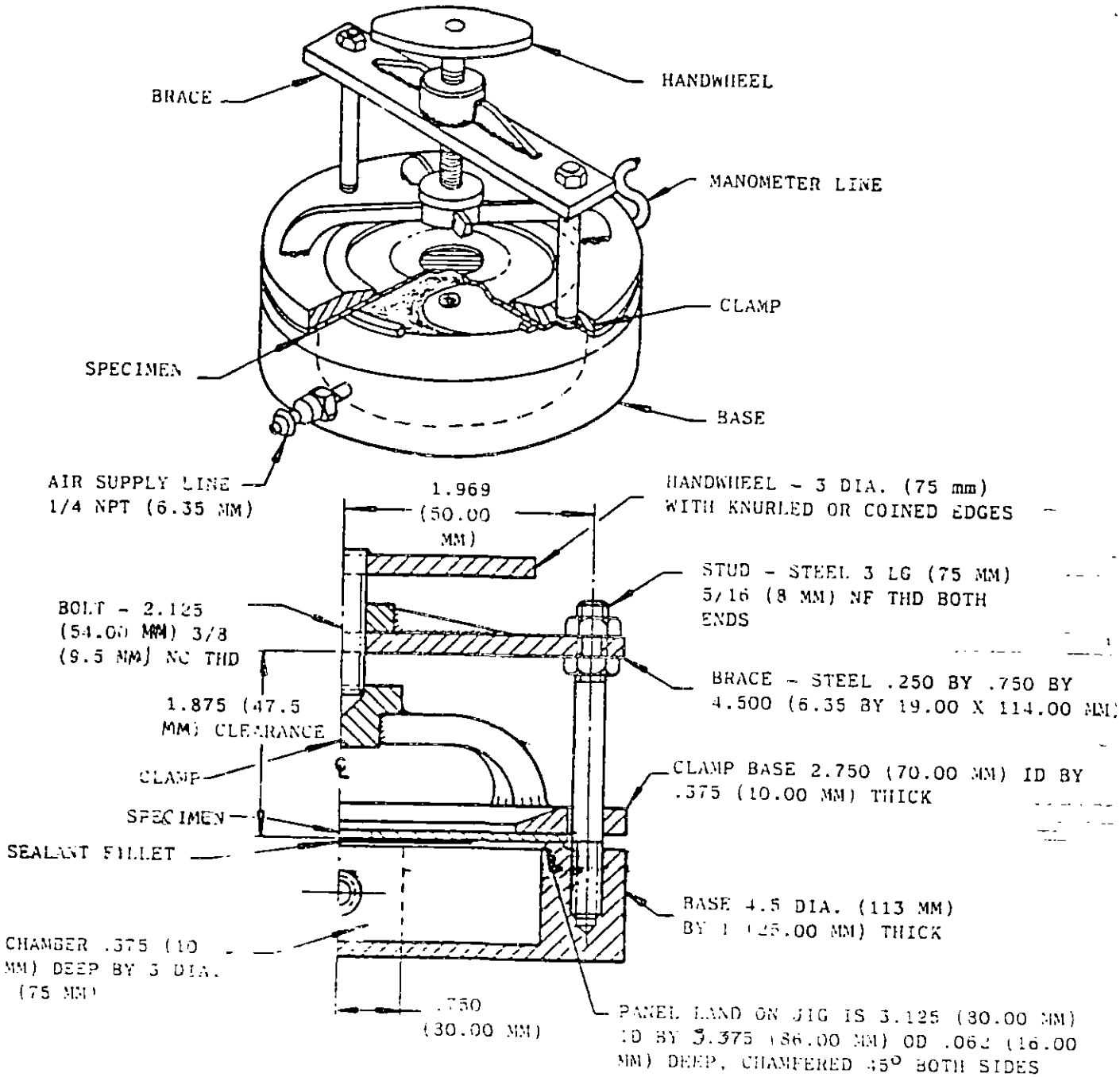
Federal Test Method Standard No. 601 after the sealing compound is allowed to cure as specified in 3.3.9. The reading shall be taken on a doubled back-to-back, 0.125 inch thick specimen.

4.8.10 Fluid immersed curing rate (classes A-1/2, B-1/4, B-1/2, and C-1/2 only). An aluminum test panel conforming to temper T6 of QQ-A-250/13 and measuring 0.040 by 2-3/4 by 6 inches (1.00 by 70 by 150 mm) shall be cleaned in accordance with 4.6.1 and covered with sealing compound to a depth of 1/4 inch (6.3 mm) in one application. After curing at standard conditions for 6 hours, the test panel shall be immersed in jet reference fluid (see 4.7) at 77°± 2°F (25°± 1°C). The hardness shall be determined after a total of 48 hours (42 hours in fluid) and after a total of 120 hours (114 hours in fluid) in accordance with method 3021 of Federal Test Method Standard No. 601.

4.8.11 Resistance to thermal rupture. Two specimens shall be prepared, each having a fillet 1/8 inch (3 mm) by 2 inches (50 mm) in diameter applied to a test panel. The test panels shall be temper T81 of QQ-A-250/4 and shall be 0.040 by 3-1/2 by 3-1/2 inches (1.00 by 90 by 90 mm) in size with a hole 1/4 inch (6 mm) in diameter in the center of the panel. The sealant fillets shall be cured for 14 days ± 4 hours at standard conditions. One of the panels shall then be placed in jet reference fluid for 120 ± 4 hours at 140°± 2°F (60°± 1°C) followed by 60 ± 4 hours at 160°± 2°F (71°± 1°C) and 6 ± 1 hour at 180°± 2°F (82°± 1°C). The panel shall be removed from the fluid and immediately applied, using a suitable gasket, to the jig shown in Figure 2. The panel shall be so positioned on the jig that the sealant is within the jig chamber. The jig shall be placed in an oven at 300°± 5°F (149°± 3°C). Using dry air, immediately apply 10 psi air pressure using an air regulator to maintain a constant pressure. The clamp jig shall be maintained in the oven for 30 ± 2 minutes after the pressure is applied. With the clamp jig still in the oven and the pressure still applied, deformation shall be measured from the surface of the test panel not exposed to pressure to the point of maximum deformation of the sealant. Repeat the test on the panel that was not immersed in jet reference fluid. Both panels shall meet the requirements of 3.3.11.

4.8.12 Low-temperature flexibility. Four test panels 0.040 inch by 2-3/4 inches by 6 inches (1.00 by 70 by 150 mm) in size shall be prepared from aluminum alloy conforming to temper T81 of QQ-A-250/4. A coating of the sealing compound 0.094 inch (2.4 mm) thick by 1-1/2 inches (40 mm) wide by 4 inches (100 mm) shall be applied to the center of each of the four panels. Care shall be taken to maintain an accurate sample thickness of 0.094 inch (2.4 mm). Panels shall be cured for 14 days ± 4 hours at standard conditions (see 4.5.1). At the end of the cure the specimens shall be immersed in 900 ml of jet reference fluid for 120 ± 4 hours at 140°± 2°F (60°± 1°C) followed by 60 ± 4 hours at 160°± 2°F (71°± 1°C) and 6 ± 1 hour at 180°± 2°F (82°± 1°C). At the completion of the fluid exposure, the specimens shall be removed

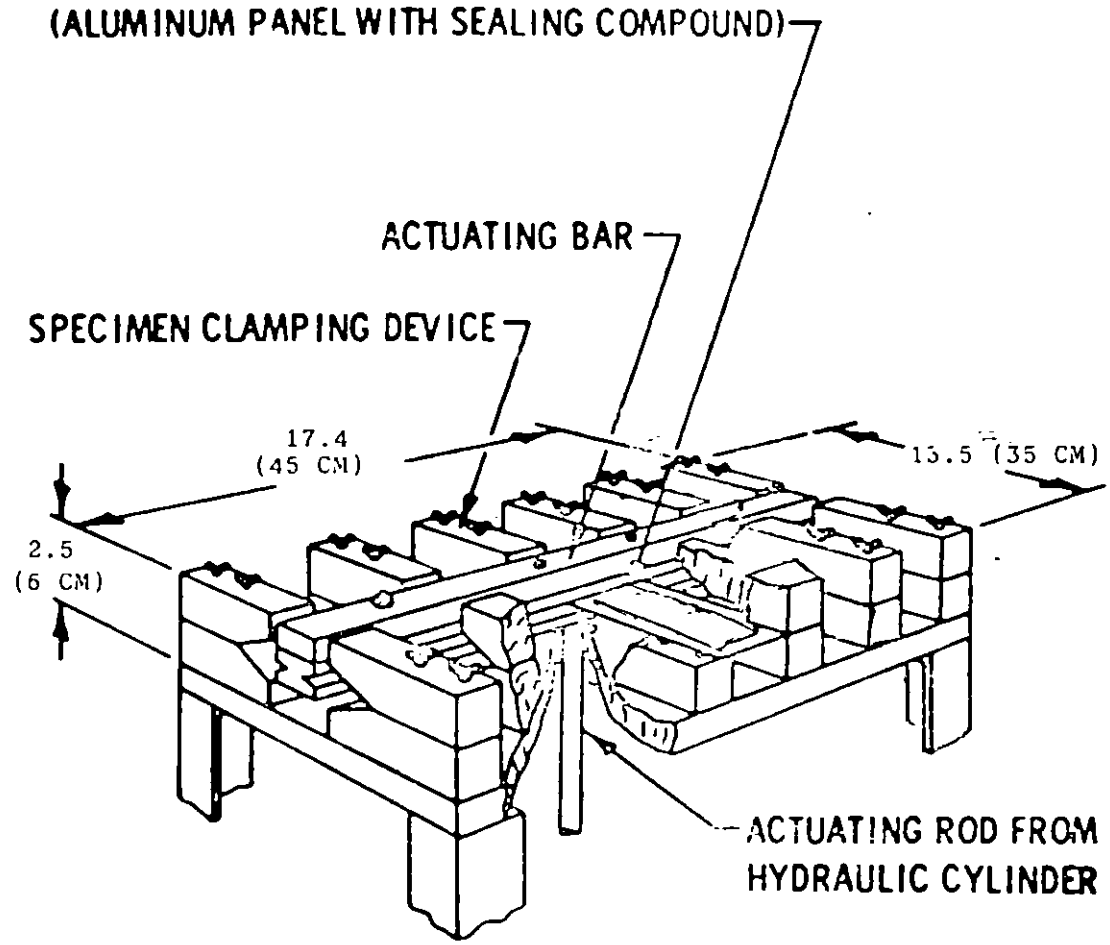
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DIMENSIONS IN INCHES (MILLIMETERS), UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ± .016 (± 0.40 MM) DEGREES ± 1

FIGURE 2. Thermal rupture jig.

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DIMENSIONS IN INCHES (CENTIMETERS).

FIGURE 3. Low temperature flexibility jig.

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from the fluid and given a standard heat cycle (4.5.2). The specimens shall then be placed in a jig which shall consist of a clamp support which will grip both sides of both six-inch (150 mm) edges of the panels for a distance of three inches (75 mm) from one end without touching the sealant. The jig shall be capable of flexing the panel through a 30 degree arc (15 degrees each side of center) at a constant frequency of 12 cycles per minute. See Figure 3. The temperature of the specimen shall be reduced to -67° ± 2°F (-55° ± 1°C) and stabilized at that temperature for a period of two hours. After stabilization, the panels shall be flexed through 130 consecutive cycles. Panels shall be removed and evaluated in accordance with 3.3.12.

4.8.13 Hydrolytic stability. A cured specimen of approximately 1/2 inch by 3 inches in diameter, shall be exposed for 120 days ± 4 hours in an environment of 160° ± 2°F (71° ± 1°C) and 95 percent relative humidity. At the expiration of this period the specimen shall be held at standard conditions (4.5.1) for 14 days and the instantaneous hardness determined.

4.8.14 Shear strength (class C-6 and C-20 only). Six test panels 0.040 by 1 by 3 inches (1.00 by 25 by 75 mm) in size shall be prepared from aluminum alloy conforming to temper T6 of QQ-A-250/13. Apply a coat of sealant 0.010 to 0.020 inch (0.25 to 0.50 mm) thick to one end of three panels covering approximately 1 inch (2.5 cm) on each panel. Overlap the sealant with another panel making a 1 square inch (6.45 cm^2) lap-shear test specimen. Reduce the thickness of the sealant to 0.005 to 0.010 inch (0.13 to 0.25 mm). Cure the sealant as per 4.6.5 and determine the shear strength by pulling in shear at a speed of 2 inches (50 mm) per minute.

4.8.15 Corrosion. One aluminum panel conforming to QQ-A-250/4 and measuring 0.040 inch by 2-3/4 inch by 6 inches (1.00 by 70 by 150 mm) in size shall be prepared with 2 parallel sealing compound fillets approximately 1 inch (2.5 cm) apart. The fillets shall be applied 3/16 inch thick by 3/4 inch wide by 5 inches long (5 mm by 2 cm by 12 cm) and shall extend to within 1/2 inch (1 cm) of the edges of the panel. The panel shall have a 14-day cure (4.5.1). At least 48 hours before the cure time has expired, the panel shall be dipped in a sealing compound topcoat conforming to MIL-S-4383 and the cure allowed to continue. At the end of the 14-day curing period, the fillets shall be cut down to 1/16 inch (2 mm) and immersed vertically for 20 days in a covered glass vessel containing a 2-layer liquid consisting of a 3 percent aqueous sodium chloride solution and jet reference fluid with 1-5/8 inches (4 cm) exposed to the salt mixture, 1-5/8 inches (4 cm) exposed to the jet reference fluid, and the remainder of the panel exposed to the air-vapor mixture. The temperature during the test shall be maintained at 140° ± 2°F (60° ± 1°C). Immediately upon removal from the liquid, the sealant shall be removed by mechanical means using a plastic scraper and the panel shall be examined for compliance with the requirements of 3.3.15.

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4.8.16 Radiographic density.

4.8.16.1 Preparation of test panels. A 6-inch (52 mm) square plate, 1/4 inch (6 mm) thick, shall be prepared from temper T81 of QQ-A-250/4. A notch 1/4 inch (6 mm) wide shall be milled to a depth of 1/8 inch (3 mm) half way across the plate. A continuation of this notch shall be milled completely through the remaining half so as to form a slot in the plate.

4.8.16.2 Application of sealant. A sample of the sealant to be tested shall be machine mixed, after which a strip 1-inch wide and 1/8 inch (3 mm) thick shall be applied over the entire length of the notched portion and slot in the test plate. A mold shall be used during the application of the sealant to the plate to assure uniform thickness of the sealant.

4.8.16.3 Test procedure. The test plate prepared in accordance with 4.8.16.1 and 4.8.16.2 shall be radiographed in accordance with MIL-STD-453 to obtain a 2 percent sensitivity through the plate at an H & D density of 2.5 $\pm$ 0.2, using Dupont 510 or Kodak M film, or equal. All density measurements shall be an average of 5 readings. Density readings shall be evaluated as specified in 3.3.16.

4.8.17 Peel strength. The panels listed in Table VI shall be used for evaluation of peel strength. At least 5 inches (12 cm) of the panels shall be coated on one side with a 1/8 $\pm$ 1/64-inch (3 $\pm$ 0.5 mm) thickness of sealing compound. A 2-3/4 by 12 inch (7 by 30 cm) strip of wire screen (20- to 40- mesh aluminum or monel wire fabric) or cotton duck conforming to Type III of CCC-CC-419 shall be impregnated with sealant, so that approximately 5 inches (12 cm) at one end is completely covered on both sides. The sealant must be worked well into the fabric. The sealant-impregnated end of the fabric shall be placed on the sealant coated panel, and smoothed down on the layer of the sealant, taking care not to trap air beneath the fabric. An additional 1/32 inch (1 mm) thick coating of sealing compound shall be applied over the fabric. After cure for 14 days $\pm$ 4 hours at standard conditions (4.5.1), the panel shall be completely immersed in covered glass vessels in the fluids and under the conditions listed in Table VI. (Immersion in wide-mouth quart jars with 2 panels in each jar is suitable.) After specified exposure at 140° $\pm$ 2°F (60° $\pm$ 1°C), the panels shall be cooled in the fluid for 24 hours at standard conditions (4.5.1). The peel strength shall be measured within 10 minutes after removal from the test fluid. Two 1-inch (2.5 cm) wide sections shall be cut through the fabric and sealing compound on each panel. The specimens shall be stripped back at an angle of 180 degrees to the metal panel in a suitable tensile testing machine having a jaw separation rate of 2 inches (50 mm) per minute. During the peel strength testing, three cuts shall be made through the sealing compound to the panel in an attempt to promote adhesive failure. The cuts shall be at

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approximate 1-inch (2.5 cm) intervals. The results shall be the numerical average of the peak loads. Failures of the sealing compound to the fabric shall not be included in the peel strength values.

4.8.17.1 Peel strength (class B-1/4, A-1/2, B-1/2, and C-1/2 only). Two 0.40 inch by 2-3/4 inches by 6 inches (1.00 by 70 by 150 mm) aluminum alloy panels which conform to QQ-A-250/12, with temper T6, shall be used. These panels shall be sulfuric acid anodized in accordance with MIL-A-8625, Type II, Class 1 (dichromate sealed) and coated with material conforming to MIL-C-27725. Sealing compound shall be applied according to 4.8.17. After curing at standard conditions for 10 hours, the test panels shall be immersed in jet reference fluid (see 4.7) at $77^{\circ} \pm 2^{\circ}$F ($25^{\circ} \pm 1^{\circ}$C) for 7 days. The panels shall then be tested according to 4.8.17.

TABLE VI. Peel strength panels.

Quantity Required	Panel	Immersion medium at $140^{\circ} \pm 2^{\circ}$F ($60^{\circ} \pm 1^{\circ}$C)
4	0.040 by 2-3/4 by 6 inches (1.00 by 70 by 150 mm) aluminum alloy, QQ-A-250/13, T6	2 panels into jet reference fluid for 7 days. 2 panels into equal parts jet reference fluid and 3% aqueous sodium chloride solution for 7 days.
4	0.040 by 2-3/4 by 6 inches (1.00 by 70 by 150 mm) aluminum alloy, QQ-A-250/12, T6 chemical treated in accordance with MIL-C-5541 (see 4.6.6)	2 panels into jet reference fluid for 7 days. 2 panels into jet reference fluid/salt water for 7 days.
4	0.040 by 2-3/4 by 6 inches (1.00 by 70 by 150 mm) aluminum alloy, QQ-A-250/12, T6 sulfuric acid anodized in accordance with MIL-A-8625, Type II, Class 1 (dichromate sealed)	2 panels into jet reference fluid for 7 days. 2 panels into jet reference fluid/salt water for 7 days.

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TABLE IV. Peel strength panels (Contd).

Quantity Required	Panel	Immersion medium at 140° ± 2°F (60° ± 1°C)
4	0.025 to 0.040 by 2-3/4 by 6 inches (0.6 to 1.00 by 70 by 150 mm), stainless steel, MIL-S-5059, composition 304, annealed, finish 2B.	2 panels into jet reference fluid for 7 days. 2 panels into jet reference fluid/salt water for 7 days.
8	0.025 to 0.040 by 2-3/4 by 6 inches (0.6 to 1.00 by 70 by 150 mm), titanium, MIL-T-9046, Type III, Composition C.	2 panels into jet reference fluid for 7 days. 2 panels into equal parts jet reference fluid and 3% aqueous sodium chloride solution for 7 days. 2 panels into jet reference fluid for 70 days with fluid change every 14 days. 2 panels into jet reference fluid/salt water for 70 days with fluid change every 14 days.
8	0.040 inch by 2-3/4 inches by 6 inches (1.00 by 70 by 150 mm) aluminum alloy, QQ-A-250/12, T6 sulfuric acid anodized in accordance with MIL-A-8625, Type II, Class I (dichromate sealed), coated with material conforming to MIL-C-27725.	2 panels into jet reference fluid for 7 days. 2 panels into equal parts jet reference fluid and 3% aqueous sodium chloride solution for 7 days. 2 panels into jet reference fluid for 70 days with fluid change every 14 days. 2 panels into jet reference fluid/salt water for 70 days with fluid change every 14 days.

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TABLE VI. Peel strength panels (Contd).

Quantity Required	Panel	Immersion medium at 140° ± 2°F (60° ± 1°C)
4	0.040 inch by 2-3/4 inches by 6 inches (1.00 by 70 by 150 mm), aluminum alloy, QQ-A-250/12, T6 sulfuric acid anodized in accordance with MIL-A-8625, Type II, Class 1 (dichromate sealed), coated with material conforming to MIL-C-27725. Before sealant application, topcoat shall be cleaned as per 4.6.1 and adhesion promoter PR-148 be applied as per 4.6.2.	2 panels into jet reference fluid for 7 days. 2 panels into jet reference fluid/salt water for 7 days.
8	0.040 inch by 2-3/4 inches by 6 inches (1.00 by 70 by 150 mm) aluminum alloy, QQ-A-250/12, T6 chemical treated in accordance with MIL-C-5541 coated with MIL-P-23377 primer. 4 panels cured 7 days at standard conditions. 4 panels cured 2 hours at 200°F (95°C).	4 panels into salt water for 7 days. 2 panels R.T. cured, 2 panels 200°F (95°C) cured. 4 panels into distilled water for 7 days. 2 panels R.T. cured, 2 panels 200°F (95°C) cured.

4.8.18 Resistance to hydrocarbons.

4.8.18.1 Weight loss and flexibility. Four 1/8 inch by 1 inch by 5 inch (3 by 25 by 125 mm) specimens shall be cut from a sheet of the sealing compound that has been cured for 14 days (4.5.1). The specimens shall be weighed and immersed in 900 cubic centimeters (cc) of jet reference fluid at 140° ± 2°F (60° ± 1°C) in a closed container for a period of seven days. At the end of the exposure period, the specimens shall be removed from the fluid and air dried for 72 hours at 120° ± 2°F (49° ± 1°C). The specimens shall then be cooled to standard test temperature in a desiccator and weighed. After weighing, the specimens shall be bent 180 degrees over a 1/8 inch (3 mm) mandrel and examined for evidence of cracking.

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4.8.18.2 Tensile strength and elongation (class B and C, except C-20). Mixed sealing compound 0.125 inch ± 0.015 inch (3.2 ± 0.4 mm) thick shall be prepared by pressing between 2 polyethylene sheets, removing the top sheet at the end of the rated tack-free time and allowing the sealing compound to cure at standard conditions. Fifteen tensile test specimens shall be cut from the sheet using die C as specified in ASTM D412. Three specimens shall be exposed to each of the environmental conditions specified in Table III. Where fluid immersion is specified, the 3 specimens shall be immersed in 400 cc of the fluid. Specimens that are tested immediately after the fluid immersion shall be cooled for 24 hours at standard conditions while remaining in the fluid and tested within 5 minutes after removal from the fluid. Specimens that are tested immediately after oven aging shall be allowed to cool for 16 to 48 hours at standard conditions. The tensile and elongation tests shall be conducted at standard conditions in accordance with ASTM D412 at a jaw separation rate of 20 inches ± 1 inch (50 ± 2 cm) per minute.

4.8.18.3 Chalking. Eight 1/8 inch by 5 inch (3 by 125 mm) specimens shall be cut from a sheet of the sealing compound that has been cured for 14 days. Four specimens shall be suspended in a closed container with 900 cc of jet reference fluid so that the specimens are totally immersed in the fluid. The other four specimens shall be similarly suspended in 900 cc of a 2 layer liquid consisting of 3 percent aqueous sodium chloride solution and jet reference fluid in equal amounts so that approximately 2-1/2 inches (63.5 mm) of each specimen is in the salt water and 2-1/2 inches (63.5 mm) is in the jet reference fluid. The test temperature shall be 140° ± 2°F (60° ± 1°C) and the test duration shall be nine days. The fluids shall be changed six times. (This means the specimens will be exposed to seven batches of fresh fluid. Nine days are used instead of seven to make allowance for weekends.) The specimens shall be allowed to remain at standard conditions at each fluid change and after testing is completed until the fluid evaporates. (Do not blot or wipe dry.) After evaporation the specimens shall be examined for chalking.

4.8.19 Repairability (all classes except C-20). A sufficient number of 0.040 by 2-3/4 by 6 inches (1.00 by 70 by 150 mm) aluminum alloy panels which conform to QQ-A-250/13 shall be used so that there are 2 panels for each class B-2 sealing compound already qualified to this specification plus 2 panels for the material being qualified. At least 5 inches (12 cm) of the panels shall be coated on one side with a 1/8 ± 1/64 inch (3 ± 0.5 mm) thickness of sealing compound so that 2 panels are coated with each class B-2 sealing compound already qualified to this

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specification and 2 panels are coated with the sealing compound being qualified. After a 14 day cure, 1 panel of each sealing compound shall be exposed to jet reference fluid for 3 days at 140° ± 2°F (60° ± 1°C) followed by 3 days of air drying at 120° ± 2°F (49° ± 1°C), then 7 days air aging at 250° ± 2°F (121° ± 1°C). The conditioned panels and the panels that received no aging shall be tested as follows: The sealant on the panels shall be cleaned in accordance with 4.6.1, and a thickness of 1/8 inches ± 1/64 inch (3 ± 0.5 mm) of newly mixed sealing compound shall be applied over the existing compound. A peel strength panel shall then be prepared in accordance with the method specified in 4.8.17. After a 14 day cure, the specimens shall be tested in peel as specified in 4.8.17 and shall comply with 3.3.19.

4.8.20 Accelerated storage. An original, unopened, 1-quart container of base compound and an original, unopened container of the curing compound shall be stored for 14 days at 120° ± 2°F (49° ± 1°C) in a suitably ventilated oven. After cooling at standard conditions for at least 24 hours, tests shall be conducted in accordance with 4.8.5, 4.8.6, 4.8.7, 4.8.8, and 4.8.9. Four QQ-A-250/12 aluminum alloy panels shall be coated with MIL-C-27725 as per Table VI. Peel panels shall be made from them as per 4.8.17 and aged for 7 days at 140° ± 2°F (60° ± 1°C) with 2 panels immersed in jet reference fluid and 2 panels in equal parts of jet reference fluid and 3 percent aqueous sodium chloride solution. The panels shall then be tested according to 4.8.17.

4.8.21 Long term storage. Three original, unopened 1-pint (1/2 liter) kits of sealing compounds (12 fluid ounces of base compound in each kit and the appropriate amount of curing compound) shall be stored at standard conditions (see 4.5.1) for nine months. At the end of this time, tests shall be conducted in accordance with 4.8.5, 4.8.6, 4.8.7, 4.8.8, and 4.8.9.

4.8.22 Hardness. The instantaneous hardness shall be determined in accordance with Method 3021 of FED-STD-601 after the sealing compound is allowed to cure for 14 days at standard conditions. The reading shall be taken on a doubled back-to-back, 0.125 inch (3.2 mm) thick specimen making the total thickness 0.25 inch (6.3 mm).

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A and C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. The base compound and the curing compound shall be packaged in separate or sectional containers. The containers,

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closure, lining, or space fillers shall not interact physically or chemically with the contents of the containers so as to alter quantity, strength, quality or purity of the container contents. The ratio of the quantity contained in the base compound container or section to the quantity contained in the curing compound container or section shall be the same as the recommended mixing ratio of the base and curing compounds.

5.1.1.1.1 Individual containers. The container containing the appropriate ratio of the curing compound shall be packaged with the container containing the unit quantity of base compound specified (see 6.2) to form a unit of issue kit in a manner that will prevent accidental container separation or mis-identification. The base compound shall be furnished in 1/2-pint, 1-pint, 1-quart, and 1-gallon metal cans conforming to PPP-P-704, or in 55-gallon drums conforming to PPP-D-729, Type III, except that tin plate cans with paper labels may be used unless otherwise specified. The base compound cans through 1-gallon sizes shall be filled to 75 percent capacity with a volume tolerance of ± 6 percent. The air in the unfilled space shall be replaced with nitrogen gas or carbon dioxide gas immediately prior to closing the container. Five-gallon pails shall contain 5 gallons of base compound and 55-gallon drums shall contain 50 gallons of base compound. The curing compound shall be furnished in the ratio quantity required by its own unit of issue base compound quantity. It shall be packaged in glass or plastic containers of adequate commercial quality and grade for the 1/2-pint, 1-pint, 1-quart, and 1-gallon sizes. The curing compound for the 5-gallon pails shall be packaged in gallon cans conforming to Type V, Class 2, PPP-C-96 and 55-gallon drums shall be packaged in pails conforming to PPP-P-704. The curing compound glass or plastic containers shall have vertical smooth inside walls and no internal projection or lips exceeding 1/16 inch. The exterior configuration of all curing compound containers shall not preclude the container to form a kit or unit of issue. Curing compound containers shall be secured to the top of its own base compound container in such a manner as to prevent accidental separation. The curing compound containers for base compound packaged in pails or drum shall be packaged together with base compound containers in containers conforming to PPP-B-636, Type CF, Class Weather Resistant, Variety SW in a manner that will prevent movement and damage to each other. The glass jars shall be closed with enameled and further sealed with cellulose bands of commercial quality, or equivalent.

5.1.1.1.2 Sectional-type containers. The base compound and curing agent shall be furnished in sectional-type 2-1/2-ounce or 6-ounce non-metal containers, conforming to MIL-P-38714, as specified in the contract or order. The total content of base compound and curing agent contained in each sectional-type container shall be as follows:

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Size of container	Total content (base and curing)	Volume tolerance
2-1/2 ounce	2 fluid ounces	± 1/8 fluid ounce
6 ounce	3-1/2 fluid ounces	± 1/8 fluid ounce

5.1.2 Level C. The unit quantity of base compound specified (see 6.3) with its ratio of curing compound shall be packaged in accordance with the manufacturer's best commercial practice.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.3).

5.2.1 Levels A and B. The base and curing compound shall be packed in accordance with the appendix to PPP-C-96 for filled cans and MIL-P-38714 for the sectioned type containers. Pails and drums shall be palletized in accordance with MIL-STD-147.

5.2.2 Level C. The base and curing compound packaged in compliance with 5.1 as specified (see 6.3) shall be packed in a manner to insure acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or Regulations of carrier applicable to the modes of transportation.

5.3 Marking of shipment. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. Marking shall include specification number, applicable class and dash number, name of manufacturer, date of manufacture (month and year), batch number, mixing instructions, and recommended storage temperature.

6. NOTES

6.1 Intended use. The sealing compound covered by this specification is intended for sealing areas of integral fuel tanks and fuel cell cavities that may be subjected to long term service temperatures of -65° to 250°F (-54° to 121°C), and short term exposure to 360°F (82° to 182°C).

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Class of sealing compound and dash number (see 1.2 and 1.2.1).

c. Quantity desired.

d. Size of container (see 5.1.1.1.1 and 5.1.1.1.2).

e. Applicable levels of packaging and packing (see 5.1 and 5.2).

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f. Special marking if required (see 5.3).

g. Source inspection required (see 3.3.23).

6.3 Qualification. With respect to products requiring qualifications, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Air Force Wright Aeronautical Laboratories, MLSA, Wright-Patterson Air Force Base, Ohio 45433 and information pertaining to qualification of products should be obtained from that activity.

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